Bihar Engineering University, Patna

B.Tech. 8th Semester Examination, 2024

Course: B.Tech. *Time: 03 Hours*
Code: 105811 *Subject: Real Time Systems* *Full Marks: 70*

Instructions: -

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / write the answer of the following (Any seven): **[2 x 7 = 14]**

(a) Real time systems must have _____.
(i) non-preemptive kernels
(ii) preemptive kernels
(iii) both of the above
(iv) none of the above

(b) The characteristic features of a hard real time system include _____
(i) embedded structure
(ii) feedback structure
(iii) distributive structure
(iv) all of the above

(c) _____ is an optimal dynamic priority scheduling algorithm.
(i) earliest deadline first
(ii) rate monotonic analysis
(iii) both of the above
(iv) none of the above

(d) Which type of constraints in real time system are imposed on the response of the controlling computer system.
(i) performance constraints
(ii) behavioural constraints
(iii) both of the above
(iv) none of the above

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(e) A system of independent, preemptable periodic tasks with relative deadlines longer than their periods can be feasibly scheduled on a processor as long as the total utilization is _____.
(i) > 1
(ii) 0
(iii) <=1
(iv) 2

(f) Priority inheritance protocol suffers from _____.
(i) deadlock
(ii) chain blocking
(iii) both of the above
(iv) none of the above

(g) _____ is based on greedy approach.
(i) priority ceiling protocol
(ii) priority inheritance protocol
(iii) both of the above
(iv) none of the above

(h) The amount of time required for the scheduling dispatcher to stop one process and start another is known as _____.
(i) dispatch latency
(ii) event latency
(iii) interrupt latency
(iv) none of the above

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(i) Which one of the following is a real time operating system?
(i) RTLinux
(ii) VxWorks
(iii) Windows CE
(iv) All of the above

(j) Task allocation of scheduling real time tasks in multiprocessor is a _____ problem
(i) P
(ii) NP
(iii) NP-hard
(iv) none of the above

Q.2 (a) Discuss any seven important characteristics of real time systems. [7]

(b) Identify the key differences between hard real time system, soft real time system, and firm real time system. Give at least one example of real time tasks corresponding to these three categories. [7]

Q.3 (a) Write different applications of real time system in the areas of industry, medical, telecommunication and aerospace. Give at least one example in each area. [7]

(b) Consider the following set of periodic real-time tasks to be scheduled by a cyclic scheduler: $T_1 = (e_1=1, p_1=4)$, $T_2 = (e_2=2, p_2=5)$, $T_3 = (e_3=5, p_3=20)$, where e_i and p_i are execution time and period of task T_i. Determine a suitable frame size for the task set. [7]

Q.4 (a) What is the working principle of time sliced round robin scheduling? Explain it using a suitable example. [7]

(b) Explain table-driven scheduler using a suitable example. [7]

Q.5 (a) Check whether the following set of three periodic real-time tasks is schedulable under Rate Monotonic Algorithm (RMA) on a uniprocessor:
$T_1 = (e_1=20, p_1=100)$, $T_2 = (e_2=30, p_2=150)$, $T_3 = (e_3=90, p_3=200)$, where e_i and p_i are execution time and period of task T_i. [7]

(b) Write down the shortcomings of Earliest Deadline First (EDF) scheduling. [7]

Q.6 (a) Describe the operations of deferrable servers in detail. [7]

(b) Discuss the operations of sporadic server in fixed-priority systems. [7]

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Q.7 (a) Explain the operations of Priority Ceiling Protocol (PCP) in sharing critical resources among real time tasks. [7]

(b) Explain how PCP is able to avoid deadlock, unbounded priority inversions, and chain blockings. [7]

Q.8 (a) Discuss any two algorithms used for allocating real time tasks on the processors of a multiprocessor system. [7]

(b) Write important features of a real time operating system. [7]

Q.9 Write Short notes on ***any two*** of the following **[7 x 2=14]**

(a) IPS protocol
(b) Priority Ceiling Protocol (PCP)
(c) Rate Monotonic Algorithm (RMA)
(d) Weighted fair queueing server

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